

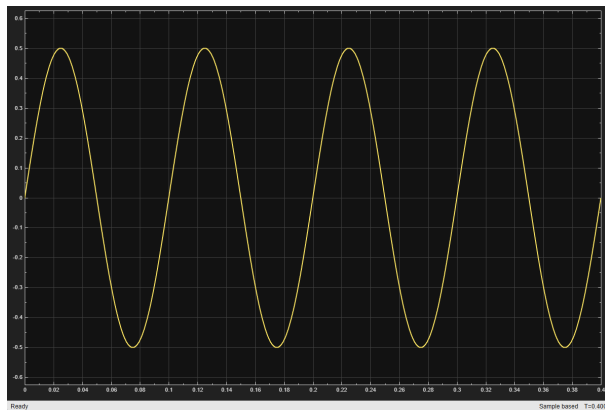
ECSE 308 - Lab 1: Signals and Noise

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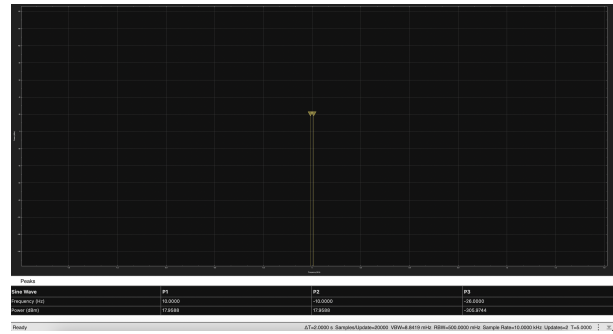
September 2026

Part 1

Question 1



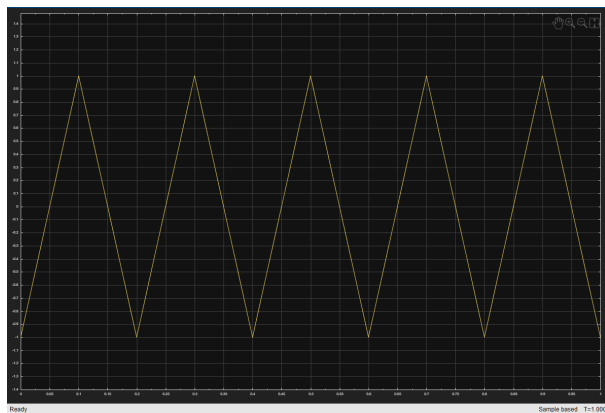
(a) Time domain.



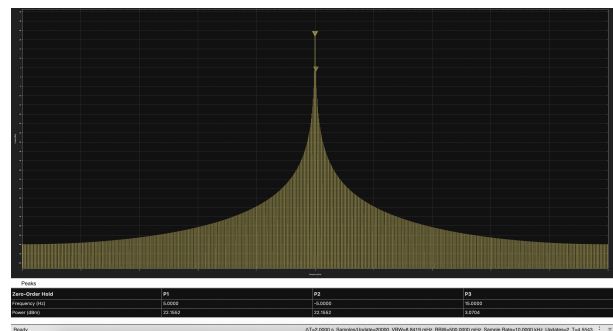
(b) Spectrum.

Figure 1: Sine wave (amplitude 0.5, $T = 0.1$ s, $f = 10$ Hz) shown over four periods, with its spectrum. An ideal sine wave does not have harmonics, but we can observe the power of 17.9588 dBm at 10 Hz matches expected amplitude of 0.5.

Question 2



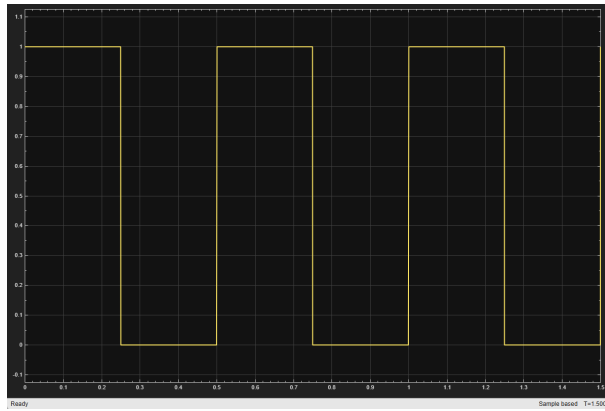
(a) Time domain.



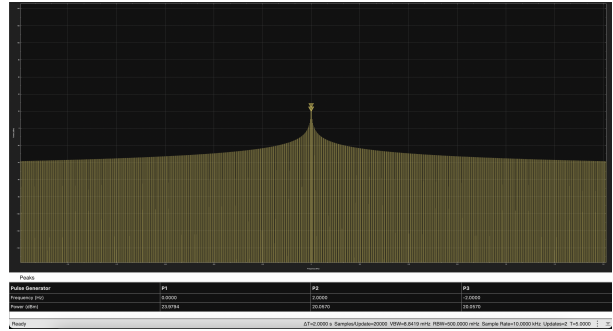
(b) Spectrum.

Figure 2: Triangular wave: amplitude 1, $T = 0.2$ s, $f_0 = 5$ Hz. Only odd harmonics (5, 15, 25, ... Hz) occur, with amplitudes decreasing as $1/n^2$.

Question 3



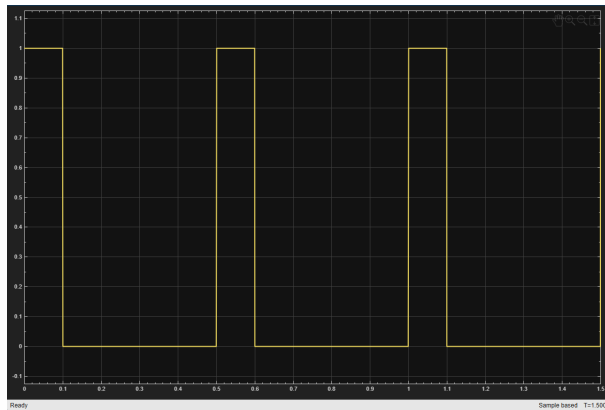
(a) Time domain.



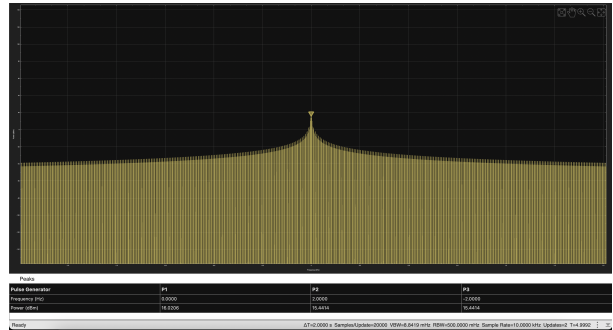
(b) Spectrum.

Figure 3: 50% duty-cycle pulse train (0 to 1.5): $T = 0.5$ s, $f_0 = 2$ Hz, DC level 0.5. Only odd harmonics (2, 6, 10, ... Hz) occur, with amplitudes decreasing as $1/n$.

Question 4



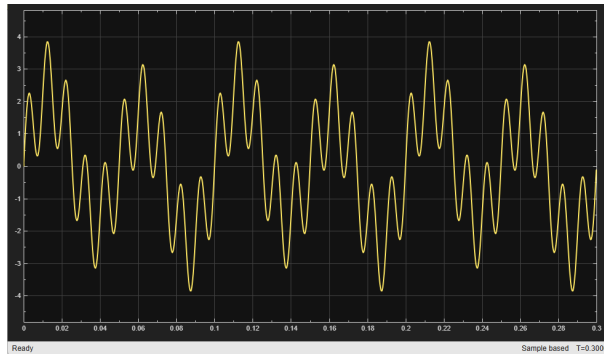
(a) Time domain.



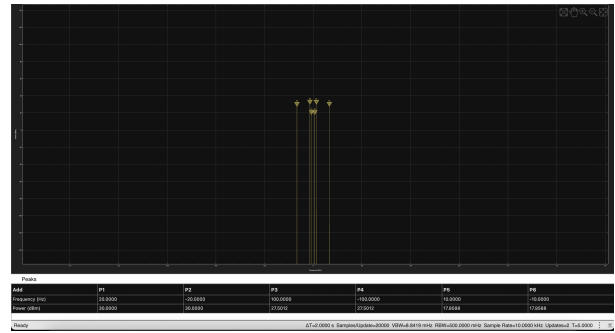
(b) Spectrum.

Figure 4: 20% duty-cycle pulse train (0 to 1.5): $T = 0.5$ s, $f_0 = 2$ Hz, DC level 0.2. Harmonics occur at $2n$ Hz, except every fifth harmonic (10, 20, 30, ... Hz), which vanishes.

Question 5



(a) Time domain.



(b) Spectrum.

Figure 5: Sum of sine waves at 10, 20 and 100 Hz with amplitudes 0.5, 2 and 1.5, respectively. The sum has $T = 0.1$ s and $f_0 = 10$ Hz; only the fundamental, second and tenth harmonics are present.

Question 6

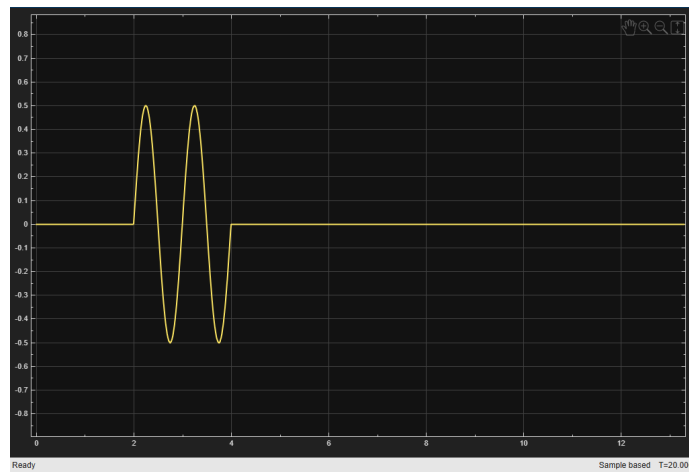
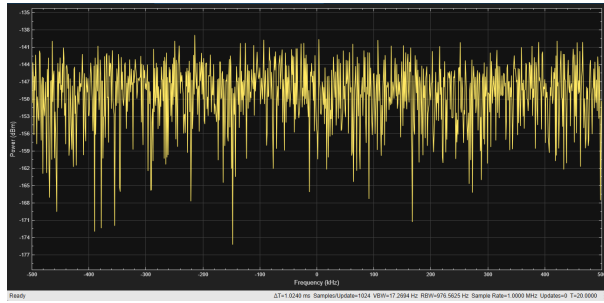
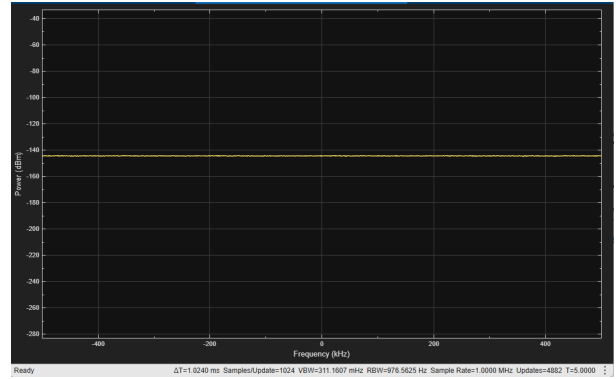


Figure 6: Difference of two step-gated sine waves (steps at $t = 2$ s and $t = 4$ s) observed on the Scope. Here we observe no overall periodicity in the resulting signal due to it being "turned off" at 4 s.

Question 7

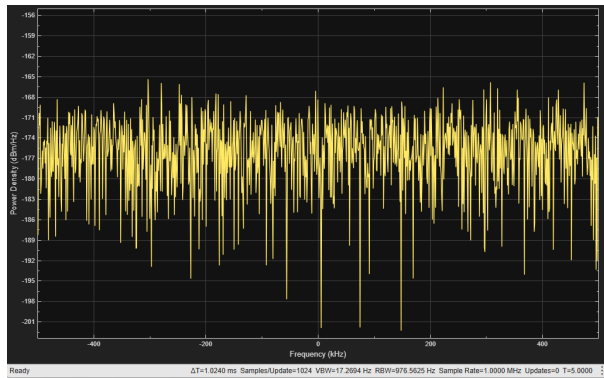


(a) Single trace.

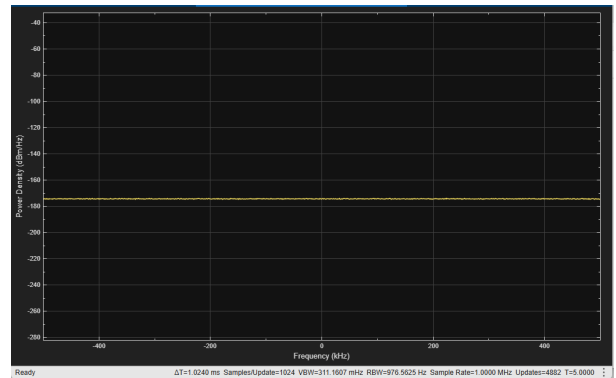


(b) With trace averaging.

Figure 7: Thermal noise at 290 K occupies the full displayed range from -500 to $+500$ kHz, a two-sided observed bandwidth of 1 MHz set by the sampling rate.



(a) Single trace.



(b) With trace averaging.

Figure 8: The thermal-noise PSD is approximately flat at -174 dBm/Hz, consistent with kT at 290 K. Trace averaging reduces fluctuations in the estimate, revealing the flat noise density.

Question 8

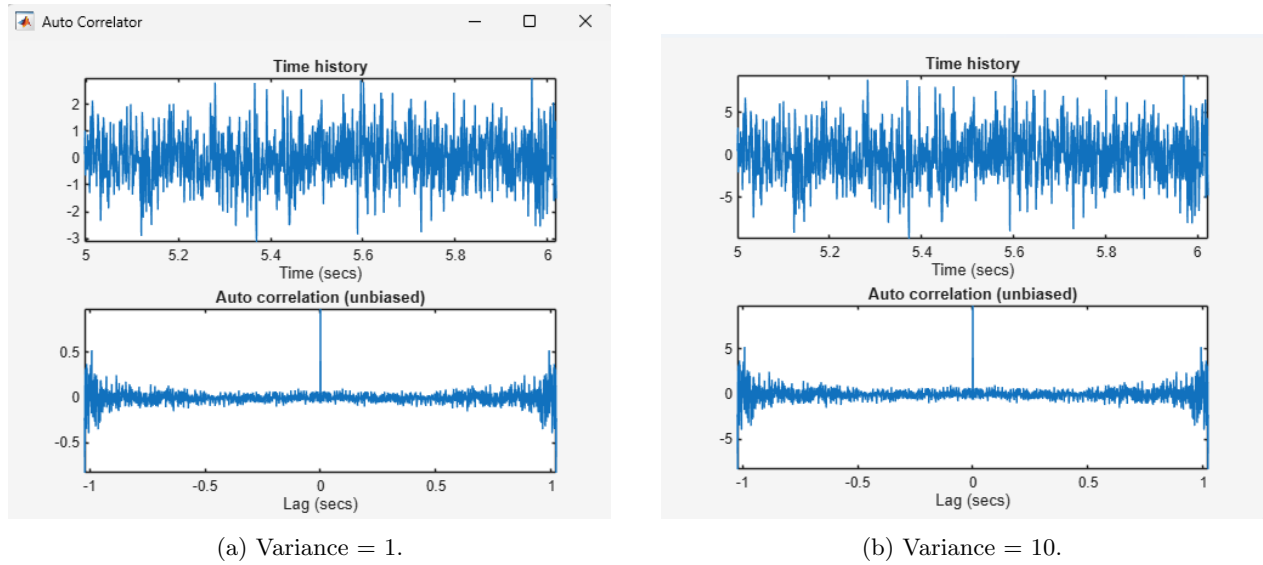


Figure 9: Autocorrelation of a zero-mean Gaussian source for two different variances. If we take the autocorrelation to be $R_{xx}(\tau) = \mathbb{E}(x(t)x(t+\tau))$, and we take the lag $\tau = 0$, we have $R_{xx}(0) = \mathbb{E}(x^2(t)) = \sigma^2$, the variance of the signal. The peak of the autocorrelation at zero lag is therefore equal to the variance, which is also the noise power for a zero-mean Gaussian source.

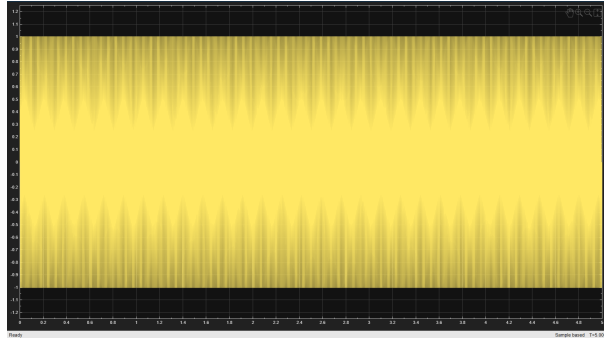
Question 9

A deterministic signal is completely specified by a mathematical expression, so its value at any time can be calculated exactly. For example, $x(t) = A \sin(2\pi ft + \phi)$ defines a sine wave once its amplitude, frequency, and phase are known.

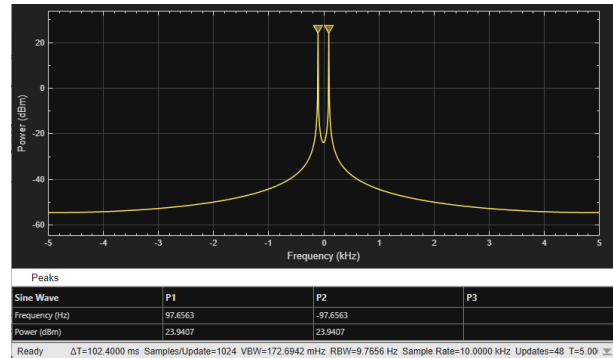
A random signal is modeled as a stochastic process: its value at each time is a random variable, and each observed waveform is one realization of that process. Rather than predicting exact values, we characterize it statistically using probability distributions, mean, variance, autocorrelation, and power spectral density.

Part 2

Question 1



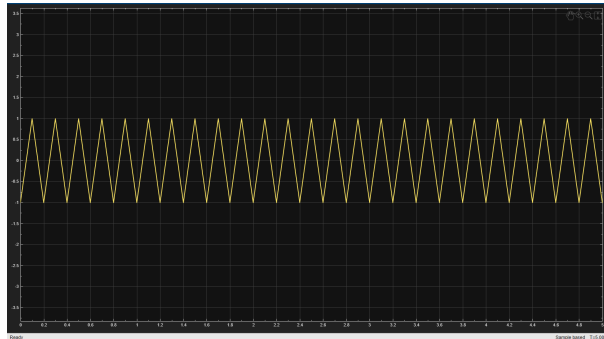
(a) Time domain.



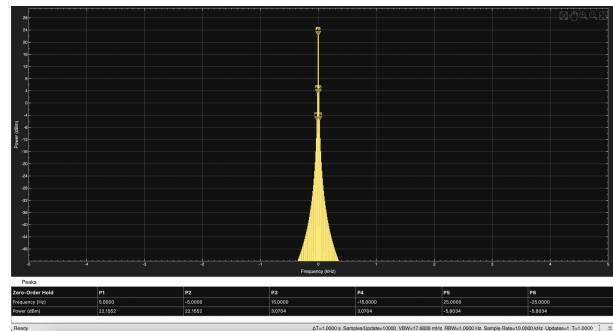
(b) Spectrum.

Figure 10: Sine wave source (100 samples per period, $T_s = 10^{-4}$ s, i.e. 100 Hz). The bandwidth should be 0 Hz for a pure sine wave, but it appears to be non-zero here due to finite FFT resolution. The total power is about $23.94\text{dBm} \times 2 = 26.94\text{dBm}$.

Question 2



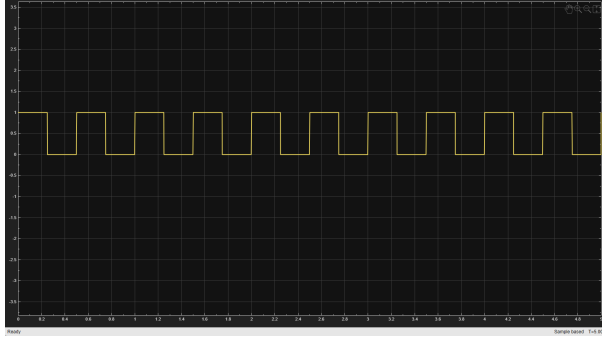
(a) Time domain.



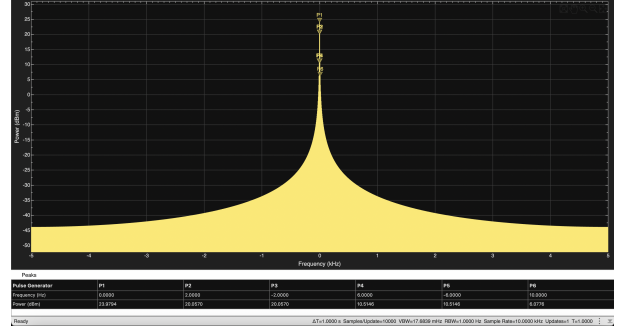
(b) Spectrum.

Figure 11: Triangular wave source at 5 Hz ($T_s = 10^{-4}$ s). The bandwidth of a triangle wave is theoretically infinite, but we do not observe this due to finite FFT resolution. The estimated total power is the sum of the power of the 3 first harmonics: $P_{total} = 22.1552 + 3.0704 - 5.8034 = 22.21\text{dBm}$.

Question 3



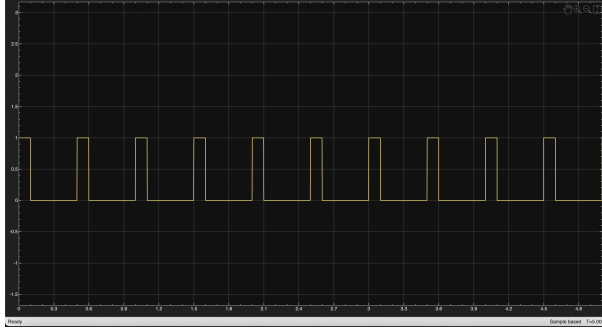
(a) Time domain.



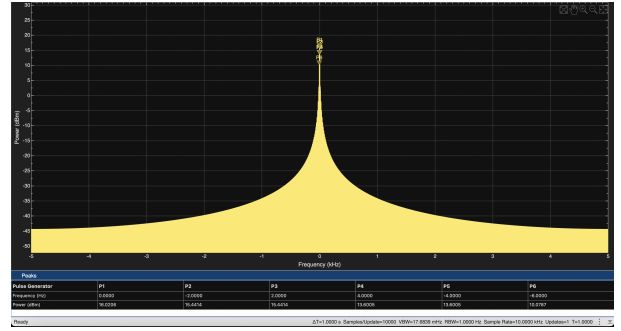
(b) Spectrum.

Figure 12: 50% duty cycle square wave source (period 5000 samples, $T_s = 10^{-4}$ s, i.e. 2 Hz). The bandwidth of a square wave is theoretically infinite, but we do not observe this due to finite FFT resolution. In our spectrum, we observe a bandwidth of 10kHz. The total power is estimated to be $P_{total} = 23.9794 + 20.0570 \times 2 + 10.5146 \times 2 = 26.77\text{dBm}$.

Question 4



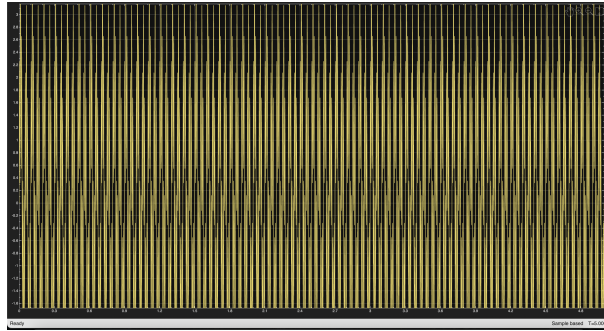
(a) Time domain.



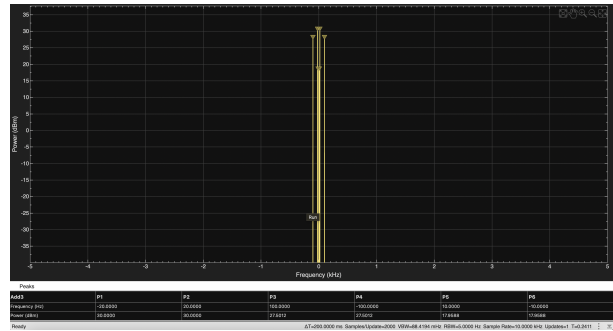
(b) Spectrum.

Figure 13: 20% duty cycle square wave source at Slider Gain 10 and $F_c = 100$ Hz. The bandwidth of a square wave is theoretically infinite, but we do not observe this due to finite FFT resolution. In our spectrum, we observe a bandwidth of 10kHz. The total power is estimated to be $P_{total} = 16.0206 + 15.4414 \times 2 + 13.6005 \times 2 = 21.93\text{dBm}$.

Question 5



(a) Time domain.

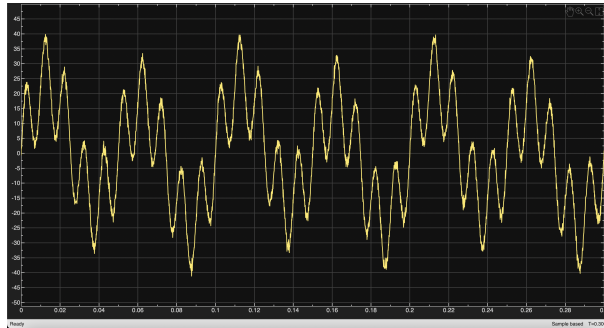


(b) Spectrum.

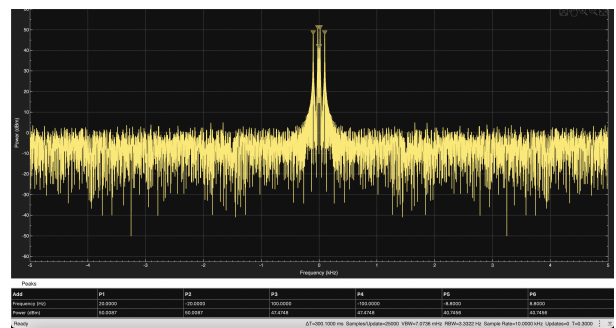
Figure 14: Sum-of-sines source at Slider Gain 10 and $F_c = 100$ Hz. The theoretical bandwidth is twice the maximum frequency of the sine waves. In this case, it's

Question 6

Adding noise to the original signal has changed both the time domain and the frequency domain signal by: making the curves noisy in time and by introducing random power in non-fundamental frequencies as well as some extra power in the fundamental frequencies.



(a) Sum of three sine waves, time domain.

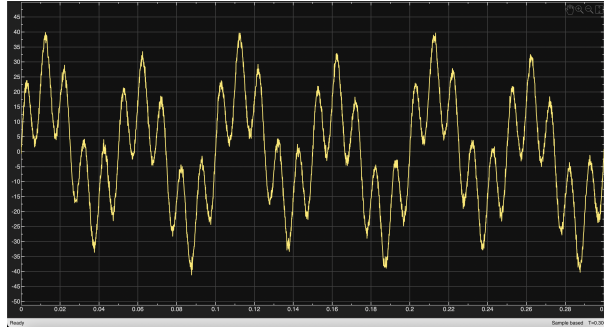


(b) Sum of three sine waves, spectrum.

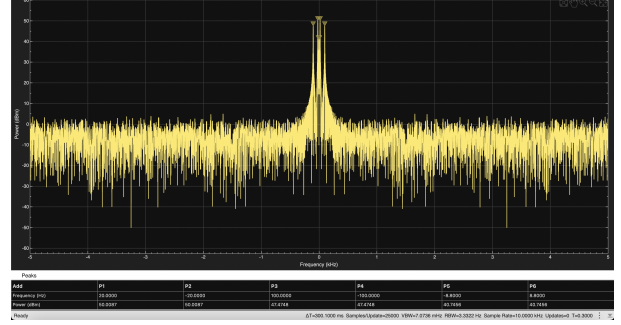
Figure 15: Received sum of three sine waves with added Gaussian noise at Slider Gain 10 and $F_c = 100$ Hz.

Question 7

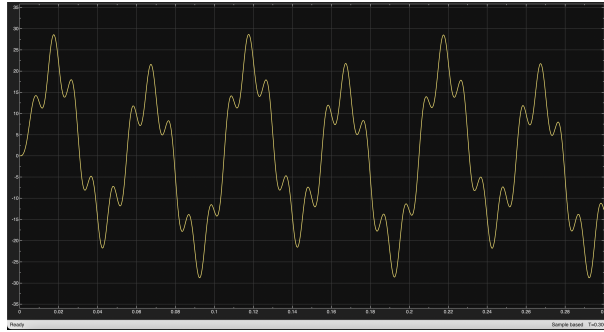
Since the filter is a low-pass filter with a cutoff frequency of 100Hz, the resulting frequency domain signal is a smoother curve with clearly pronounced peaks at the expected frequencies (those observed in question 5). This is because the filter removes the high-frequency noise (although, not all of it). The resulting time-domain signal is also a little smoother (there is still some roughness due to low-frequency noise still being present).



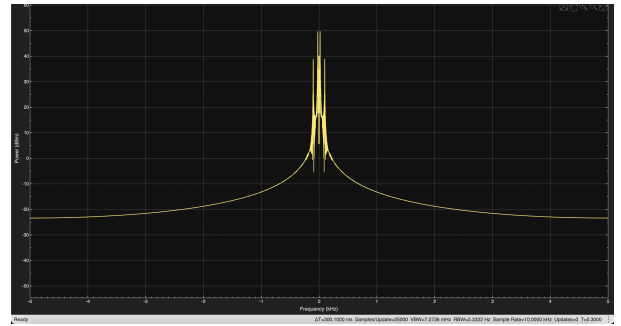
(a) Received, time domain.



(b) Received, spectrum.



(c) Filtered, time domain.



(d) Filtered, spectrum.

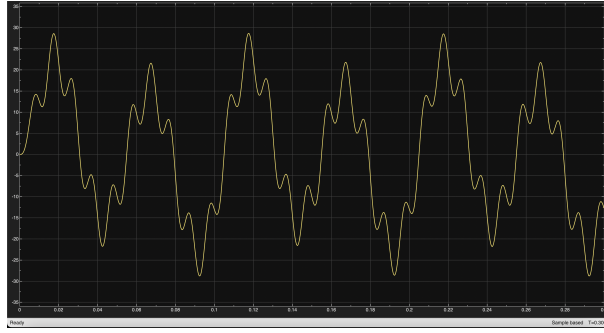
Figure 16: Sum of three sine waves before filtering (top) and after filtering (bottom), at Slider Gain 10 and $F_c = 100$ Hz.

Question 8

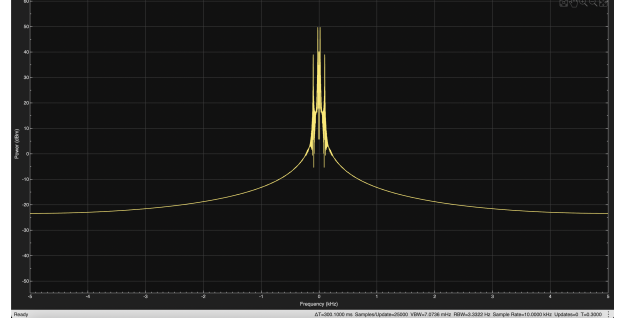
With the results below, we can observe the following:

- Increasing the signal gain means a higher SNR. A higher SNR means that the noise has less of an effect on the resulting signal, so the filtered signal appears to be less and less noisy (even though the noise itself hasn't changed). This also continues into the frequency domain, where the spectral graph is smoother with higher SNR.
- Increasing the cutoff frequency means that more noise is kept in the resulting signal. This basically means a lower SNR, which implies a noisier signal. A noisier signal also means that the frequency domain graph has a larger and larger bandwidth of just noise, which ends up making the time-domain graph noisier.

Baseline: gain 10, $F_c = 100$ Hz



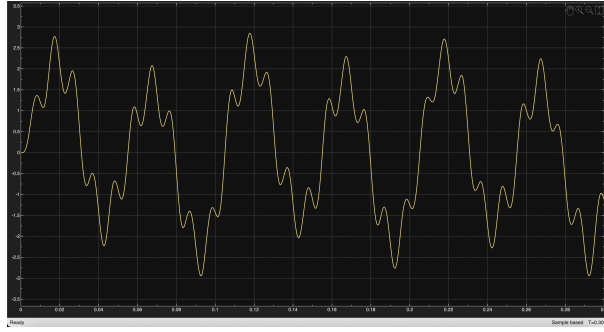
(a) Time domain.



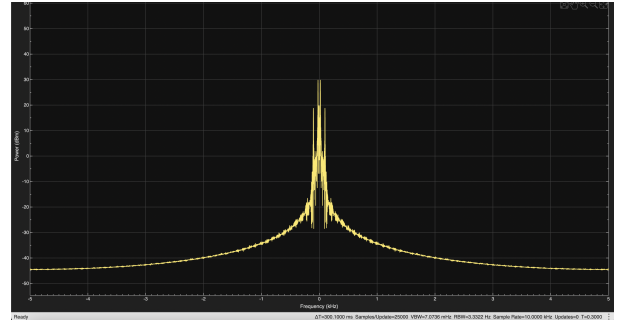
(b) Spectrum.

Figure 17: Baseline: filtered sum of three sine waves at gain 10 and $F_c = 100$ Hz.

Varying the gain ($F_c = 100$ Hz)

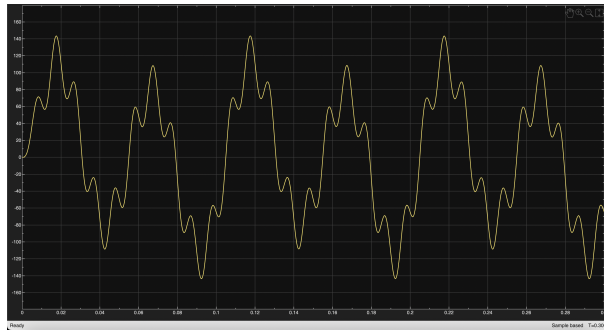


(a) Time domain.

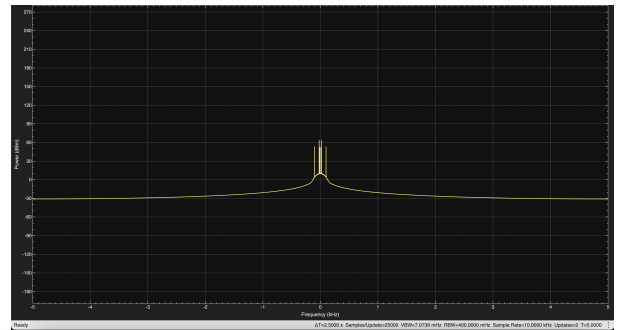


(b) Spectrum.

Figure 18: Lower gain: filtered sum of three sine waves at gain 1 and $F_c = 100$ Hz.



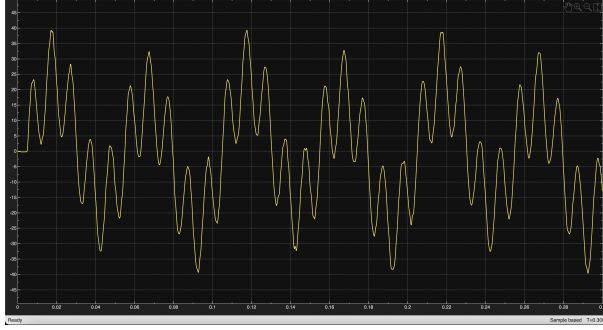
(a) Time domain.



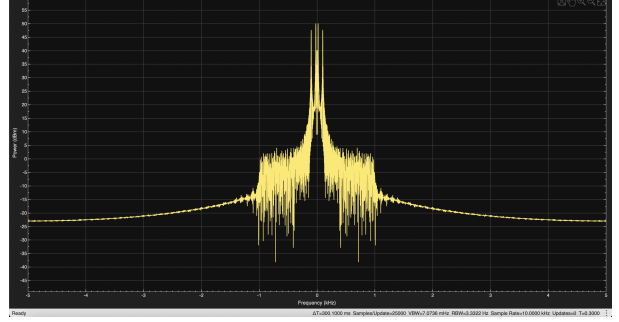
(b) Spectrum.

Figure 19: Higher gain: filtered sum of three sine waves at gain 50 and $F_c = 100$ Hz.

Varying the cutoff frequency (gain 10)

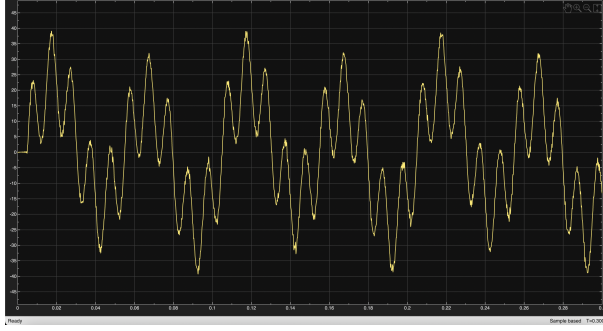


(a) Time domain.

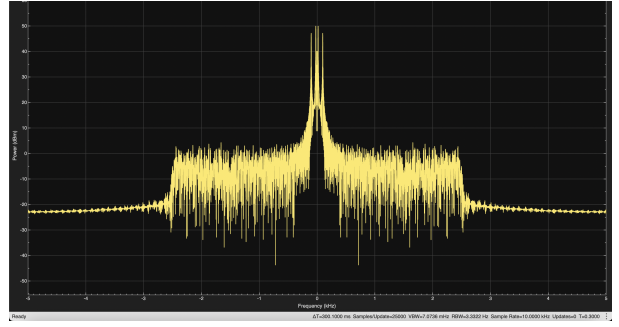


(b) Spectrum.

Figure 20: Higher cutoff frequency: filtered sum of three sine waves at gain 10 and $F_c = 1000$ Hz.



(a) Time domain.



(b) Spectrum.

Figure 21: Higher cutoff frequency: filtered sum of three sine waves at gain 10 and $F_c = 2500$ Hz.